

RISHABH GUPTA

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EDUCATION

Vellore Institute of Technology, Vellore
B.Tech in Computer Science and Engineering

August 2023 – July 2027
CGPA: 9.33 / 10

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, Java, HTML, CSS
- **Frameworks & Libraries:** React, Streamlit, Pandas, NumPy, Scikit-learn
- **Tools & Platforms:** VS Code, Arduino IDE, GitHub, Docker, AWS, Microsoft Azure, Power BI

RESEARCH AND PUBLICATIONS

AI-Powered Adaptive MCQ Generation Based on Bloom's Taxonomy and Real-Time Student Performance

Under Review - Discover Journal

- Conducted a comprehensive survey of **50+** research works (2015–2025) on AI-driven adaptive assessment systems.
- Compared transformer and sequential models (BERT, GPT) for cognitive-level classification aligned with Bloom's Taxonomy.
- Analyzed components of adaptive learning systems, including LLM-based MCQ generation, knowledge tracing (DKT, SAKT), sequencing, and feedback mechanisms.
- Examined bias in AI-generated MCQs (position, similar options) and compared mitigation methods, highlighting limitations in fairness and usability.

PROJECTS

AskDrAI | Python, Flask (REST API), RAG, Google FLAN-T5, IBM Watsonx Assistant, Streamlit, Ngrok

- Built an AI-driven triage system processing **15+** symptoms, integrating RAG with FLAN-T5 to recommend departments.
- Engineered a Flask REST API to handle symptom ingestion, inference orchestration, and response delivery.
- Integrated IBM Watsonx Assistant to enable multi-turn conversational workflows, exposing services via Ngrok.
- Validated system behavior through **50+ test** interactions using a lightweight Streamlit interface for rapid iteration.

Quantitative Time-Series Modeling Engine | Python, Pandas, NumPy, Plotly, Groww API

- Developed a multi-factor "Value Cost Averaging" model and an event-driven backtesting framework.
- Built an automated pipeline processing **1,700+** time-series datasets for large-scale financial analysis.
- Engineered **5+** factor feature set (Wilder's RSI, ATR, rolling quantiles) modeling market momentum, volatility, & value.
- Evaluated model alpha by benchmarking ROI against baseline strategies, enabling data-driven tuning and strategy selection.

EXPERIENCE

Robotics Intern – Robotics R&D

June 2025 – July 2025

Armstrong Dematic

- Contributed to ROS 2 Nav2–based navigation system for AMR, covering path planning, localization, and recovery behaviors.
- Fixed critical Nav2 recovery-loop failure via **20+** simulations and real-robot tests, eliminating redundant recoveries.
- Reduced unintended mission aborts by **100%** during obstacle encounters.
- Collaborated with **5+** Robotics R&D engineers to analyze logs, validate fixes, and support autonomous operation.

LEADERSHIP & ACTIVITIES

Avionics Head

August 2025 – Present

Team Veloce

- Architected and led **autonomous drone development**, leveraging Pixhawk flight controllers and ArduPilot (Mission Planner) with integrated GPS navigation, telemetry logging, and FPV audio-video systems.
- Led a team of **4** at IIT Kanpur Techfest, building and operating a rover for navigation and task-based challenges.
- Qualified for the AIAA Design–Build–Fly (DBF) Fly-Off (2026) by securing **11th position globally** in the proposal round.
- Mentored **6 juniors** in avionics and robotics, driving hands-on execution across projects.

CERTIFICATIONS

- **Supervised Machine Learning: Regression and Classification** – Andrew Ng, Coursera
- **Microsoft Elevate Internship: Power BI** – Microsoft India, AICTE
- **GenAI using IBM Watsonx (CEWXAI1IN)** – IBM (IBMCE)